TED UNIVERSITY / SOFTWARE ENGINEERING & COMPUTER ENGINEERING

PROJECT PROPOSAL

SENG 491 / CMPE 491 - Senior Project I | Fall 2026 | 23 October 2026

Project Name: CommerceIQ - AI-Powered E-Commerce Intelligence and Decision Support System

Project Website: https://commerceaiq-website.vercel.app/

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PROJECT DESCRIPTION

Small and medium-sized e-commerce businesses generate sales, inventory, and cost data, yet often lack the analytical tools needed to translate these records into practical decisions. Demand uncertainty can lead to stockouts or excess inventory, while fragmented information makes it difficult to assess product-level profitability. CommerceIQ is a proposed AI-powered e-commerce intelligence and decision support system intended to address these challenges by bringing operational data into a unified analytical workflow. The system aims to support sellers in understanding historical performance, anticipating future demand, and identifying inventory and profitability considerations before making purchasing and replenishment decisions.

The project will investigate suitable demand forecasting methods and develop an integrated approach that connects forecast outputs with inventory replenishment recommendations and cost-aware profitability analysis. Planned capabilities include structured data import, an analytical dashboard, interpretable forecasts, and explainable decision-support recommendations. The engineering process will involve requirements analysis, system and data modeling, implementation, testing, and evaluation using appropriate forecasting and usability measures. The intended contribution is not merely to predict sales, but to make predictive insights actionable for everyday e-commerce operations while acknowledging data quality, uncertainty, and the importance of human judgment.